be an unprecedented human catastrophe. There's something like one chance in two thousand that such a collision will happen in the lifetime of a newborn baby. Most of us would not fly in an airplane if the chance of crashing were one in two thousand. (In fact for commercial flights the chance is one in two million. Even so, many people consider this large enough to worry about, or even to take out insurance for.) When our lives are at stake, we often change our behavior to arrange more favorable odds. Those who don't tend to be no longer with us.

Perhaps we should practice getting to these worldlets and diverting their orbits, should the hour of need ever arise. Melville notwithstanding, some of the chips of creation *are* still left, and improvements evidently need to be made. Along parallel and only weakly interacting tracks, the planetary science community and the U.S. and Russian nuclear weapons laboratories, aware of the foregoing scenarios, have been pursuing these questions: how to monitor all sizable near-Earth interplanetary objects, how to characterize their physical and chemical nature, how to predict which ones may be on a future collision trajectory with Earth, and, finally, how to prevent a collision from happening.

The Russian spaceflight pioneer Konstantin Tsiolkovsky argued a century ago that there must be bodies intermediate in size between the observed large asteroids and those asteroidal fragments, the meteorites, that occasionally fall to Earth. He wrote about living on small asteroids in interplanetary space. He did not have military applications in mind. In the early 1980s, though, some in the U.S. weapons establishment argued that the Soviets might use near-Earth asteroids as first-strike weapons; the alleged plan was called "Ivan's Hammer." Countermeasures were needed. But, at the same time, it was suggested, maybe it wasn't a bad idea for the United States to learn how to use small worlds as weapons of its own. The Defense Department's Ballistic Missile Defense Organization, the successor to the Star Wars office of the 1980s, launched an innovative spacecraft called *Clementine* to orbit the Moon and fly by the near-Earth asteroid Geographos. (After completing a remarkable reconnaissance